

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **high-performance microservices** using **Node.js** and **JavaScript** backend frameworks (**Express.js**, **Fastify**) to support secure wallet infrastructure and digital asset operations, delivering **sub-100ms API latency** across distributed systems handling **10K+ concurrent transactions**.
- Built **scalable distributed systems** and **peer-to-peer technologies** leveraging **microservices architecture** patterns, **event-driven design**, and **eventual consistency** models to ensure high-availability backend platforms supporting **99.99% uptime SLA** across global regions.
- Implemented backend security principles and **secure coding practices** including cryptographic asset protection, input validation, rate limiting, and API authentication (**JWT**, **OAuth 2.0**) to safeguard user assets and sensitive blockchain transaction data across wallet infrastructure.
- Engineered data persistence layers using **MySQL** and **MongoDB** to optimize transaction ledger storage, real-time balance synchronization, and distributed data retrieval across wallet microservices, reducing query latency by **40%** through schema optimization and indexing strategies.
- Collaborated with cross-functional international engineering teams using **Agile development environments** and **Git-based CI/CD pipelines** to deliver innovative backend solutions aligned with product and business requirements, conducting code reviews and architectural design sessions to improve system reliability and maintainability by **35%**.
- Investigated, troubleshooted, and resolved backend performance issues and distributed system faults using **observability tools** (**Splunk**, **CloudWatch**, **Dynatrace**) to proactively identify optimization opportunities, reducing mean-time-to-resolution for critical wallet infrastructure incidents from **45 minutes to 12 minutes**.

SKILLS

Backend & Microservices – Node.js, JavaScript, Express.js, Fastify, microservices architecture, distributed systems, peer-to-peer technologies, event-driven design, high-availability platforms, scalability optimization.

Databases & Data Systems – MySQL, MongoDB, PostgreSQL, Oracle, data modelling, distributed data synchronization, transaction ledgers, schema optimization, indexing strategies.

Security & Compliance – Secure coding practices, cryptographic asset protection, JWT, OAuth 2.0, input validation, rate limiting, API authentication, data protection, regulatory compliance.

DevOps & Infrastructure – Docker, AWS (EC2, S3, Lambda, CloudWatch), CI/CD pipelines, Jenkins, Git branching, Linux, observability (Splunk, Dynatrace), performance troubleshooting.

Blockchain & Digital Assets – Blockchain technologies, decentralized finance, digital asset ecosystems, wallet infrastructure, fintech backend systems, peer-to-peer protocols.

Development Practices – Agile development, code reviews, architectural design, REST APIs, problem-solving, performance optimization, remote team collaboration.

Supporting Technologies – Python, Java, Spring Boot, SQL, Jira, Confluence, analytics, data engineering, statistical analysis.

EDUCATION AND TRAINING

PROJECTS

Decentralized Wallet Transaction Engine (HCL Technologies)

- Architected and built a **Node.js-powered microservices backend** for a high-throughput **blockchain wallet platform**, implementing **event-sourced transaction processing** and **peer-to-peer asset settlement** protocols to handle **50,000+ digital asset transfers daily** with sub-second confirmation times.
- Designed **MySQL and MongoDB hybrid data layer** to separate transactional ledgers (MySQL) from distributed asset state (MongoDB), enabling **eventual consistency across 12 global nodes** while maintaining cryptographic proof integrity and reducing synchronization conflicts by **92%**.
- Implemented end-to-end **backend security architecture** including **JWT-based authentication**, **RSA-2048 encryption** for sensitive wallet keys, and **rate limiting** to prevent double-spend attacks, protecting **\$2.3M in customer digital assets** across production infrastructure.
- Optimized **microservices performance** using **Docker containerization**, **AWS Lambda for async settlement jobs**, and **CloudWatch metrics-driven auto-scaling**, reducing API response latency from 850ms to **120ms average** and improving throughput by **340%** during peak trading windows.
- Delivered **scalable distributed wallet state synchronization** using **event-driven architecture** and **message queues** to ensure balance consistency across **peer-to-peer node networks**, achieving **99.98% data consistency** with zero customer-reported reconciliation errors over 18 months.

SWIFT Payments Operations Dashboard & Digital Asset Reconciliation (HCL Technologies)

- Built production-grade **backend API services** supporting a SWIFT messaging and payments-operations platform, implementing **ISO 20022 message parsing**, **SEPA compliance**, and real-time transaction-flow visibility for **cross-border payment settlement** across European and global banking corridors.
- Developed **secure, scalable backend microservices** using **Java Spring Boot** and **REST APIs** to orchestrate **peer-to-peer payment routing**, regulatory audit trails, and **GDPR-compliant data handling**, reducing payment processing latency by **28%** while maintaining **100% regulatory compliance** with banking authority requirements.
- Engineered **SQL-based transaction reconciliation system** using **PostgreSQL and Oracle** to validate **ISO 20022 message flows**, detect settlement anomalies, and auto-reconcile cross-border payments, processing **2.1M transactions monthly** with **99.97% accuracy** and zero regulatory exceptions.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)